Frequency Response Measurement With Loopinator

Loopinator generates a frequency response plot by sweeping a sinusoid through a range of frequencies and measuring the magnitude and phase at the output. These instructions will guide you through setting up the equipment and Loopinator program to obtain your very own Bode plot.

1 Quick Start Steps

Following is an outline of the necessary steps to generate a Bode plot:

- **Test equipment setup (refer to Figure 1)**
  1. Attach the BNC to breadboard cable to the "Gen Out" port on the oscilloscope
  2. Attach the Channel 1 probe to measure the *INPUT* of your circuit
  3. Attach the Channel 2 probe to measure the *OUTPUT* of your circuit
- **Loopinator setup**
  1. Use NI Max to obtain the VISA resources for the oscilloscope (Figure 2)
  2. In Loopinator, set the VISA resource for the oscilloscope (Figure 4)
  3. Set the desired oscilloscope settings (Figure 5)
  4. Set up the desired wave generator parameters (Figure 6)
  5. Set the desired start and stop frequency and points per decade (Figure 6)

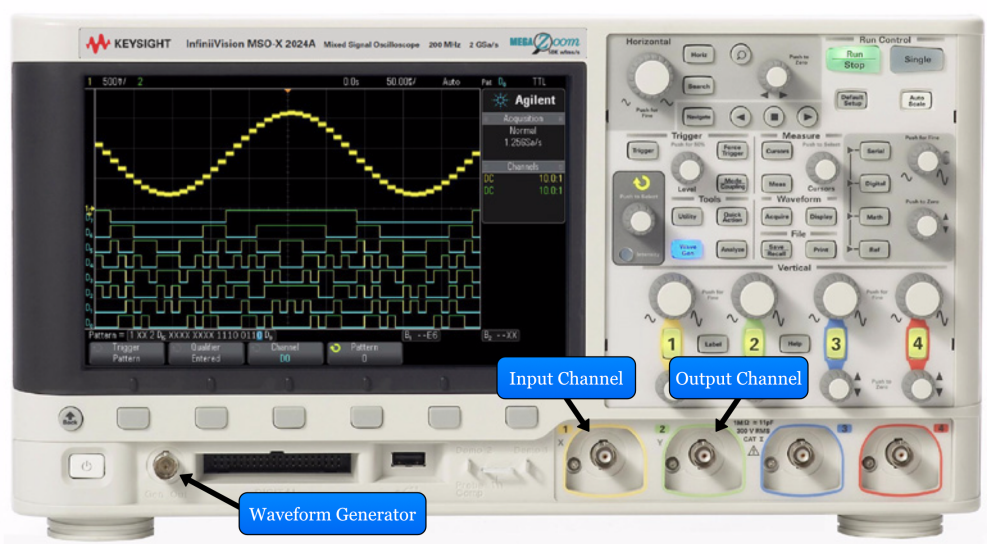

Figure 1: Ports for connecting oscilloscope to circuit

2 Loopinator Interface

In order to communicate with the test equipment, Loopinator needs to know the VISA[1] resource address for the oscilloscope (obtained through NI Max). Once the VISA address is entered into Loopinator, you can set up your desired parameters and hit "START."

2.1 Getting the VISA Resource From NI Max

1. Open Windows the start menu and search for "NI Max".
2. In NI Max locate the VISA resource address for the oscilloscope (Figure 2). Keep this window open as a reference to the VISA resources.

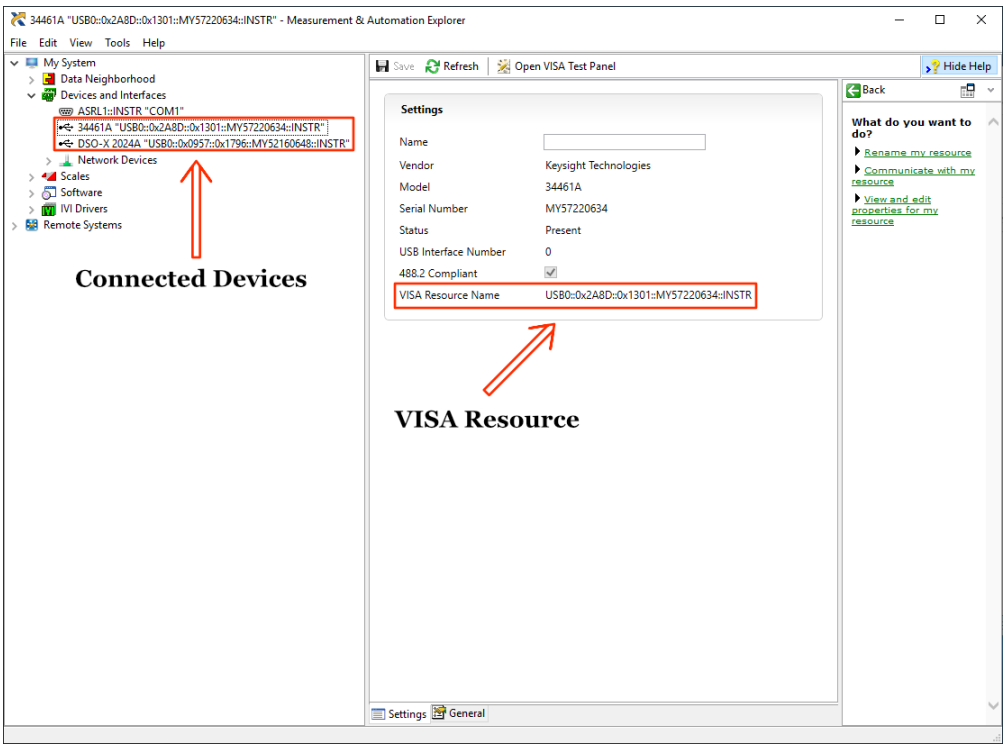

Figure 2: NI Max VISA addresses

[1]Virtual instrument software architecture, or VISA, is an industry standard API for communicating with electrical test equipment from a computer. The VISA resource is how the computer identifies the particular instrument.

2.2 Setting up The Loopinator

Open Loopinator by searching it in the start menu. You will be greeted with the Loopinator "Front Panel" (Figure 3). This is where you will set all the instrument and frequency sweep parameters.

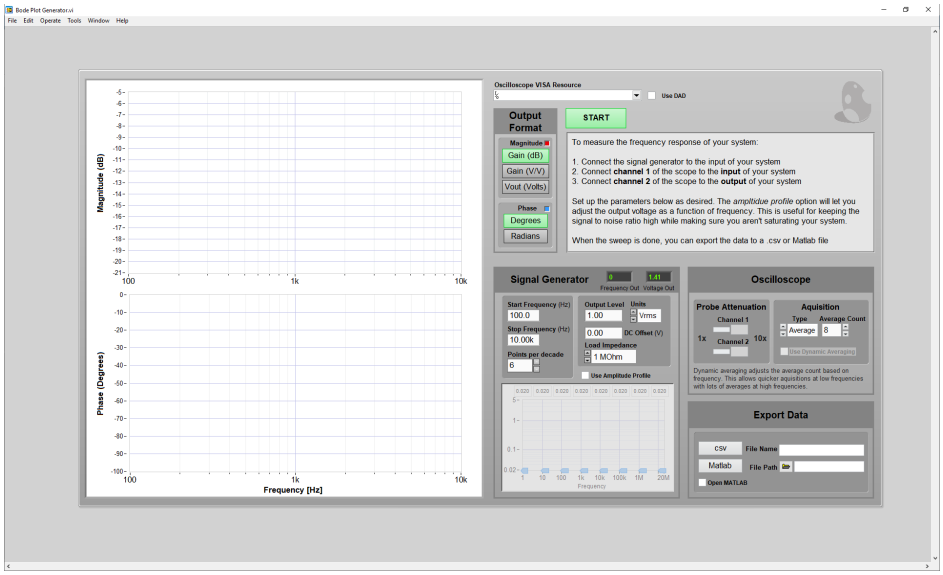

Figure 3: Loopinator Front Panel

1. Use the drop down menu to select the VISA resource addresses indicated in NI Max. If you're using the Digilent Analog Discover (DAD), check the "Use DAD" checkbox. (Figure 4)

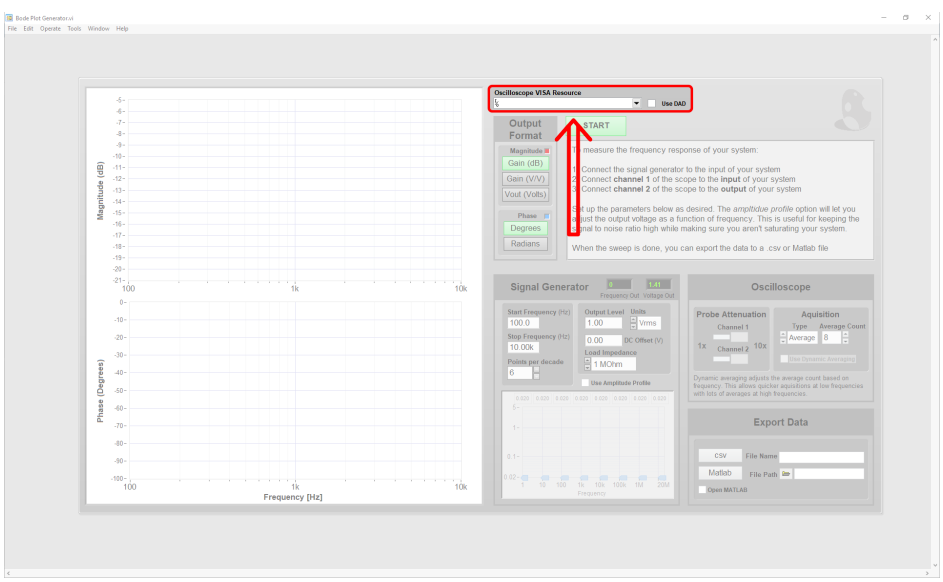

Figure 4: Location of VISA resource input on Front Panel

2. Set up the oscilloscope channels. Note that Channel 1 is the ***input*** and Channel 2 is the ***output*** of your circuit. Set the probe attenuation to match the probes you're using–usually 10×.[2] (Figure 5)

   - The Acquisition mode options are the same ones available on the Keysight oscilloscope. Usually the best option for these kind of sweeps is "Average", but this can be slow if you're measuring low frequencies (less than 100 Hz or so), so you may want to lower the average count or use a different acquisition mode in this case.
   - The scope will automatically adjust the Volts/div scale as it sweeps to maximize the capture resolution.[3].

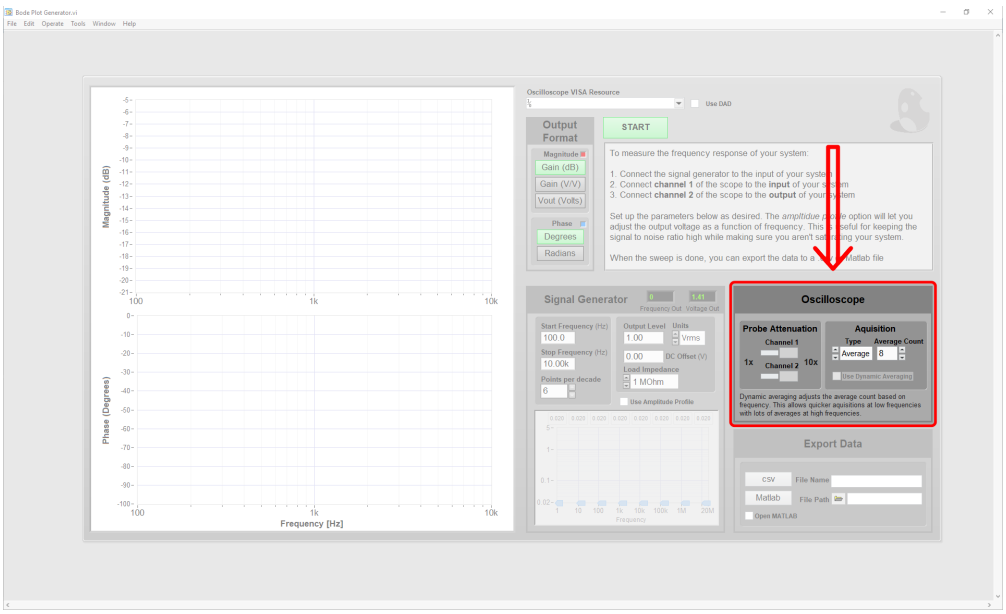

Figure 5: Oscilloscope setup

[2]Because the transfer function is a ratio $\frac{V_o}{V_i}$, it's not crucial that these are set, but if you want to measure the raw voltages you will need to make sure these match your probes

[3]The 'scope uses an 8-bit ADC (analog to digital converter) to capture the waveform. This means that the vertical resolution of the waveform is equal to $\frac{vertical\ range}{255\ bits}$ where $vertical\ range = (volts/div) * (10\ divisions)$. In other words, the more of the screen you use to view the wave, the greater resolution you will get.

3. Set up the signal generator by inputting your voltage, volt units, start frequency, stop frequency, and number of points per decade. (Figure 6)

   - It is advised that you use a large v_{in} (max is 5 v_{pp}) to get better readings of the attenuated signals (i.e., improve your signal to noise ratio or SNR).
   - If you need a DC Offset you can set that here, but note that the sweeps are captured with AC coupling turned on, so you won't be able to see the DC value on the oscilloscope.
   - The "Amplitude Profile" option allows you to dynamically change the the output voltage as a function of frequency.[4]

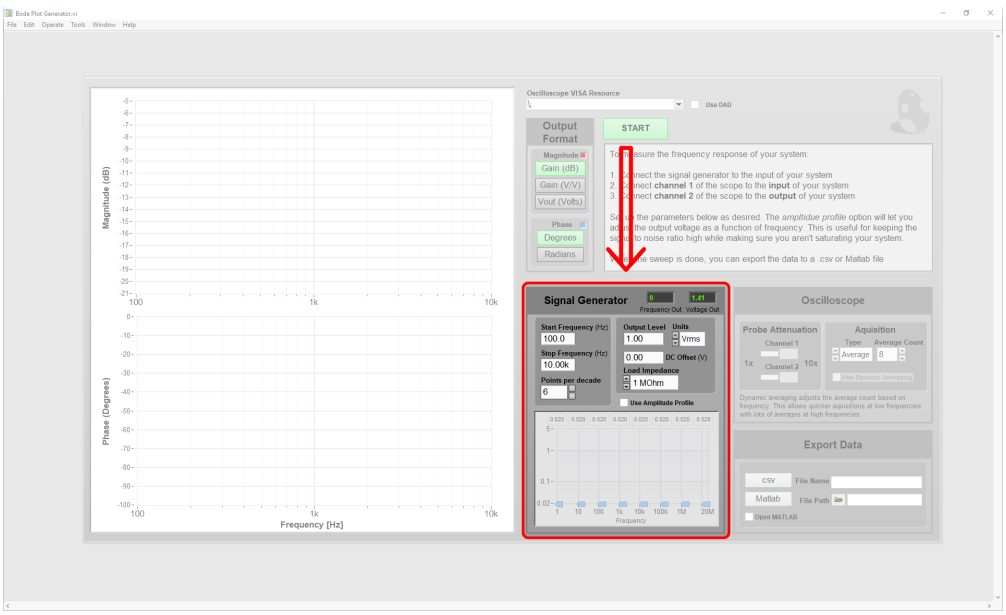

Figure 6: Signal generator parameters

[4]This is useful for maximizing SNR in systems where the input voltage has different limits at different frequencies. For example, when doing power supply open-loop response measurements.

4. Choose the output plot units. (Figure 7)

   - **Gain (dB)** plots $20\log_{10}\left(\frac{V_o}{V_i}\right)$
   - **Gain (V/V)** plots $\frac{V_o}{V_i}$
   - **Vout (Volts)** plots just V_o
   - Note the STOP button will halt the program but it can be slow. Once you hit it, wait for the scope's Run/Stop button to be red before restarting.

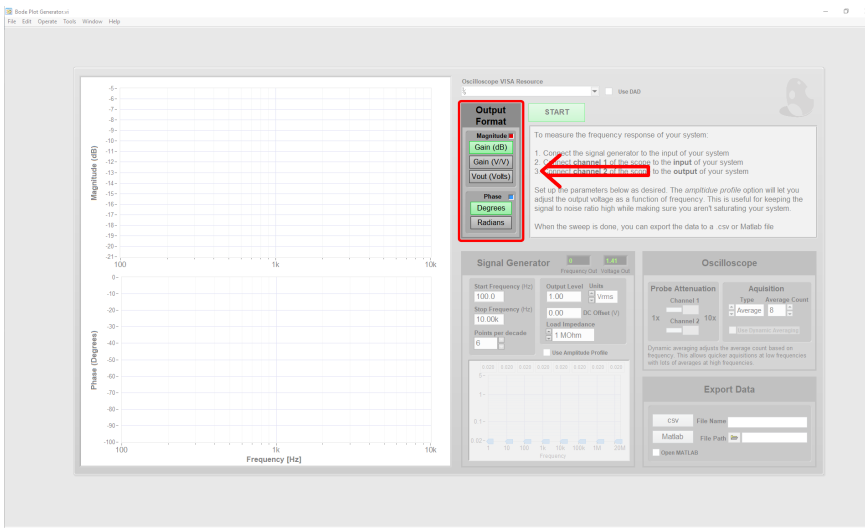

Figure 7: Output display format

5. Click the START button to begin the sweep. As it runs, you should see the plot being drawn. (Figure 8)

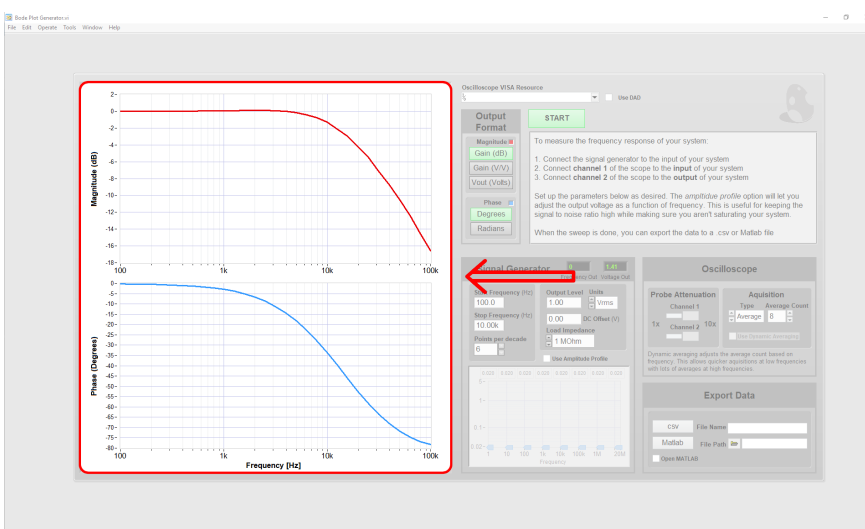

Figure 8: Live plotted output

6. Export your data to a `.csv` or Matlab file.

   - Put the file name in the "File Name" box
   - browse to the desired save path by clicking the folder icon
   - Clicking the "csv" button will save the output data as a `.csv`
   - Clicking the "Matlab" button will save the data as a `.m` that can be ran in Matlab to import the data
   - Checking the "Open MATLAB" checkbox will cause the `.m` to be opened directly in Matlab

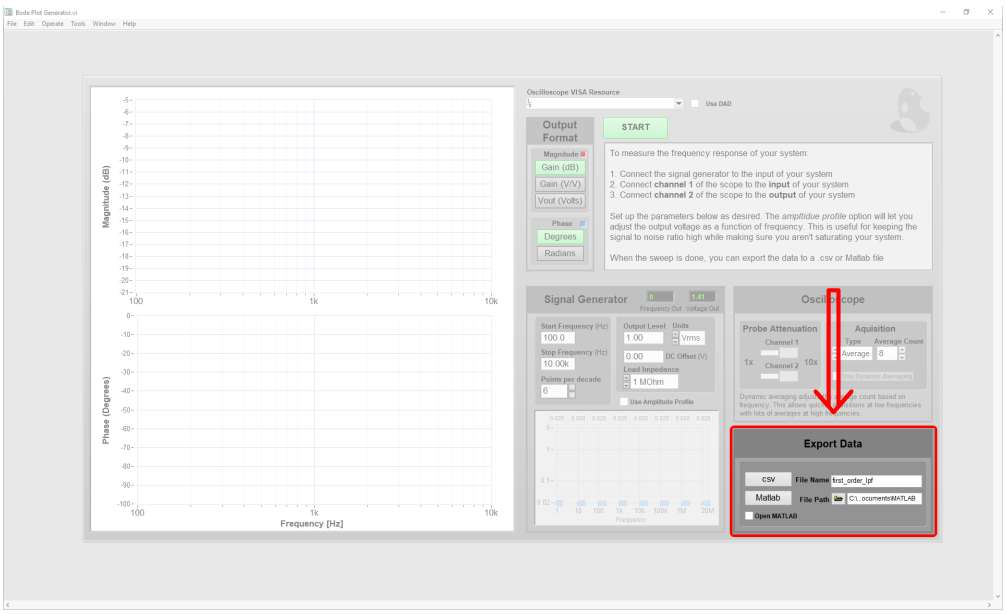

Figure 9: Export options

3 Common Problems

As with all test equipment, there are problems and limitations. Here are some common things to check if you're having trouble:

Common Problems

- **You are receiving a VISA Open error**
  - This usually occurs because an instrument is not found or a VISA resource was not properly closed. Check you VISA Resource address against the ones listed in NI Max. If they look correct, restart the test instruments.
- **You are receiving an ID Query Failed error**
  - This usually occurs because the wrong VISA resource address was given to Loopinator. Double check the address in NI Max.
- **The output channel is not adjusting properly on the oscilloscope and the phase and gain plots are clearly incorrect**
  - This usually occurs because Channel 1 and Channel 2 are swapped. Channel 1 is the *input* to the circuit; Channel 2 is the *output* of the circuit.

Limitations

- Stopping the frequency sweep can be slow using the STOP button while it finishes working on whatever it was doing when a stop was requested and closes the VISA resource–be patient. Aborting the program with the top left tool-bar stop button causes the VISA resource to remain open and could disrupt the connection to the instrument. Always use the STOP button when possible.
- The DAD doesn't have all the features of the Keysight scopes, especially with relation to the Acquisition mode. The future of this program likely lies in Python so many of the features will be added when it gets ported (if you're interested in this kind of thing, contact ETG!)